

OCR Further Mathematics
Pure Core - Further Algebra
Question Paper
AS and A Level
Further Mathematics A - H235, H245



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The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.

- 1(a) A function $f(z)$ is defined on all complex numbers z by $f(z) = z^3 - 3z^2 + kz - 5$ where k is a real constant. The roots of the equation $f(z) = 0$ are α , β and γ . You are given that $\alpha^2 + \beta^2 + \gamma^2 = -5$.

Explain why $f(z) = 0$ has only one real root.

[3]

- (b) Find the value of k .

[3]

- (c)

Find a cubic equation with integer coefficients that has roots $\frac{1}{\alpha}$, $\frac{1}{\beta}$ and $\frac{1}{\gamma}$.

[2]

- 2(a) In this question you must show detailed reasoning.

Find the values of A , B and C for which $\frac{x^3 + x^2 + 9x - 1}{x^3 + x^2 + 4x + 4} \equiv A + \frac{Bx + C}{x^3 + x^2 + 4x + 4}$.

[1]

- (b)

Hence express $\frac{x^3 + x^2 + 9x - 1}{x^3 + x^2 + 4x + 4}$ using partial fractions.

[5]

- (c)

Using your answer to part (b), determine $\int_0^2 \frac{x^3 + x^2 + 9x - 1}{x^3 + x^2 + 4x + 4} dx$ expressing your answer in the form $a + \ln b + c\pi$ where a is an integer, and b and c are both rational.

[4]

3(a) In this question you must show detailed reasoning.

Use partial fractions to show that $\sum_{r=5}^n \frac{3}{r^2 + r - 2} = \frac{37}{60} - \frac{1}{n} - \frac{1}{n+1} - \frac{1}{n+2}$. [5]

(b)

Write down the value of $\lim_{n \rightarrow \infty} \left(\sum_{r=5}^n \frac{3}{r^2 + r - 2} \right)$. [1]

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4(a) In this question you must show detailed reasoning.

The roots of the equation $3x^3 - 2x^2 - 5x - 4 = 0$ are α , β and γ .

Find a cubic equation with integer coefficients whose roots are α^2 , β^2 , and γ^2 .

[4]

(b)

Find the exact value of $\frac{\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2}{\alpha\beta\gamma}$.

[2]

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Show that $\int_0^{\frac{1}{\sqrt{3}}} \frac{4}{1-x^4} dx = \ln(a + \sqrt{b}) + \frac{\pi}{c}$ where a , b and c are integers to be determined.

[6]



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6(a) You are given that the cubic equation $2x^3 + px^2 + qx - 3 = 0$, where p and q are real numbers, has a complex root $\alpha = 1 + i\sqrt{2}$.

Write down a second complex root, β .

[1]



(b) Determine the third root, γ .

[2]

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(c) Find the value of p and the value of q .

[2]

(d)

Show that if n is an integer then $\alpha^n + \beta^n + \gamma^n = 2 \times 3^{\frac{1}{2}n} \times \cos n\theta + \frac{1}{2^n}$ where $\tan \theta = \sqrt{2}$.

[4]

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7 Prove by induction that the sum of the cubes of three consecutive positive integers is divisible by 9.

[5]



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8(a) The quadratic equation $x^2 - 2x + 5 = 0$ has roots α and β .

Write down the values of $\alpha + \beta$ and $\alpha\beta$.

[1]

(b)

Hence find a quadratic equation with roots $\alpha + \frac{1}{\beta}$ and $\beta + \frac{1}{\alpha}$.

[3]

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9 In this question you must show detailed reasoning.

Find $\int_2^3 \frac{x+1}{x^3-x^2+x-1} dx$, expressing your answer in the form $a \ln b$ where a and b are rational numbers.

[6]



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10 The roots of the equation $ax^2 + bx + c = 0$, where a , b and c are positive integers, are α and β .

(a) Find a quadratic equation with integer coefficients whose roots are $\alpha + \beta$ and $\alpha\beta$.

[4]



(b) Show that it is not possible for the original equation and the equation found in part (a) both to have repeated roots.

[2]

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(c) Show that the discriminant of the equation found in part (a) is always positive.

[3]



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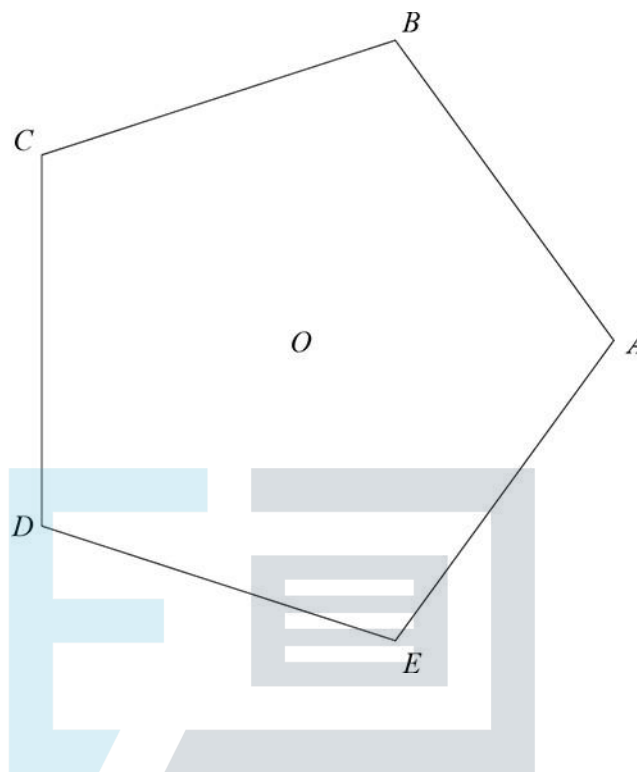
11 © You are given that the cubic equation $2x^3 - 3x^2 + x + 4 = 0$ has three roots, α , β and γ .

By making a suitable substitution to obtain a related cubic equation, determine the value of

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}.$$

[4]

- 12 The Argand diagram below shows the origin O and pentagon $ABCDE$, where A, B, C, D and E are the points that represent the complex numbers a, b, c, d and e , and where a is a positive real number. You are given that these five complex numbers are the roots of the equation $z^5 - a^5 = 0$.



(a) Justify each of the following statements.

- (i) A, B, C, D and E lie on a circle with centre O .

[1]

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© 2024 (ii) $ABCDE$ is a regular pentagon.

[2]

(iii) $b \times e^{\frac{2i\pi}{5}} = c$

[1]

(iv) $b^* = e$

[1]

(v) $a + b + c + d + e = 0$

[2]

- (b) The midpoints of sides AB , BC , CD , DE and EA represent the complex numbers p , q , r , s and t . Determine a polynomial equation, with real coefficients, that has roots p , q , r , s and t .

[3]

- 13 The equation $x^3 + 2x^2 + x + 3 = 0$ has roots α , β and γ .
The equation $x^3 + px^2 + qx + r = 0$ has roots $\alpha\beta$, $\beta\gamma$ and $\gamma\alpha$.
Find the values of p , q and r .

[5]

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- 14 It is given that $\frac{5x^2 + x + 12}{x^3 + kx} \equiv \frac{A}{x} + \frac{Bx + C}{x^2 + k}$ where k, A, B and C are positive integers.

Determine the set of possible values of k .

[5]

- 15 The roots of the equation $x^3 - kx^2 - 2 = 0$ are α, β and γ , where α is real and β and γ are complex.

- (i) Show that $k = \alpha - \frac{2}{\alpha^2}$.

[2]

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- (ii) Given that $\beta = u + iv$, where u and v are real, find u in terms of α .

[4]

- (iii) Find v^2 in terms of α .

[4]

16 The quadratic equation $x^2 + x + k = 0$ has roots α and β .

(i) Use the substitution $x = 2u + 1$ to obtain a quadratic equation in u .

[2]

(ii) Hence, or otherwise, find the value of $\left(\frac{\alpha-1}{2}\right)\left(\frac{\beta-1}{2}\right)$ in terms of k .

[2]

17

(i) Show that $(\alpha\beta + \beta\gamma + \gamma\alpha)^2 \equiv \alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 + 2\alpha\beta\gamma(\alpha + \beta + \gamma)$.

[3]

(ii) It is given that α , β and γ are the roots of the cubic equation $x^3 + px^2 - 4x + 3 = 0$,

where p is a constant. Find the value of $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$ in terms of p .

[5]

18 You are given the equation $(2x - 1)^2 - e^x = 0$.

(i) Verify that 0 is a root of the equation.

[1]

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There are also two other roots, α and β , where $0 < \alpha < \beta$.

(ii) The iterative formula $x_{r+1} = \ln(2x_r - 1)^2$ is to be used to find a root of the equation.

(a) Sketch the line $y = x$ and the curve $y = \ln(2x-1)^2$ on the same axes, showing the roots 0, α and β .



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[3]

(b) By drawing a 'staircase' diagram on your sketch, starting with a value of x that is between α and β , show that this iteration does not converge to α .

[1]

(c) Using this iterative formula with $x_1 = 3.75$, find the value of β correct to 3 decimal places.

[3]

(iii) Using the Newton-Raphson method with $x_1 = 1.6$, find the root α of the equation $(2x - 1)^2 - e^x = 0$ correct to 5 significant figures. Show the result of each iteration.

[4]



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Express $\frac{5x}{(x-1)(x^2+4)}$ in partial fractions.



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It is given that $f(x) = \frac{x^2 - 25}{(x-1)(x+2)}$.

(i) Express $f(x)$ in partial fractions.



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[4]

(ii) Write down the equations of the asymptotes of the curve $y = f(x)$.

[2]

(iii) Find the value of x where the graph of $y = f(x)$ cuts the horizontal asymptote.

[2]

(iv) Sketch the graph of $y^2 = f(x)$.

[2]

21 The quadratic equation $kx^2 + x + k = 0$ has roots α and β .

(i) Write down the values of $\alpha + \beta$ and $\alpha\beta$.

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[1]

(ii) Find the value of $\left(\alpha + \frac{1}{\alpha}\right)\left(\beta + \frac{1}{\beta}\right)$ in terms of k .

[5]



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22 The cubic equation $x^3 + 4x + 3 = 0$ has roots α , β and γ .

(i) Use the substitution $x = \sqrt{u}$ to obtain a cubic equation in u . [3]

(ii) Find the value of $\alpha^4 + \beta^4 + \gamma^4 + \alpha\beta\gamma$. [7]

23 The cubic equation $kx^3 + 6x^2 + x - 3 = 0$, where k is a non-zero constant, has roots α , β and γ .

Find the value of $(\alpha + 1)(\beta + 1) + (\beta + 1)(\gamma + 1) + (\gamma + 1)(\alpha + 1)$ in terms of k . [6]

24 The cubic equation $2x^3 + 3x + 3 = 0$ has roots α , β and γ .

(i) Use the substitution $x = u + 2$ to find a cubic equation in u .

[3]

(ii) Hence find the value of $\frac{1}{\alpha-2} + \frac{1}{\beta-2} + \frac{1}{\gamma-2}$.

[4]

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END OF QUESTION PAPER

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